Year 11 Higher

REVISION PACK

Progress Check 4

ANSWERS

Units in this pack

- Graphs 6
- Handling Data 5

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 6 · Graphs to solve equations (Lessons 8.1, 8.2)

Q	Working	Answer
solving with a quadratic graph		
1.1	(a) crosses the x-axis (b) $x^2 - 3x - 2 = -x + 2$ draw $y = -x + 2$	**(a)** $x = -0.6$ or 3.6 **(b)** $x = -1.2$ or 3.2 **(line** $y = -x + 2$**)**
1.2	(a) crosses the x-axis (b) $x^2 - x - 3 = x - 2$ draw $y = x - 2$	**(a)** $x = -1.3$ or 2.3 **(b)** $x = -0.4$ or 2.4 **(line** $y = x - 2$**)**
1.3	(a) crosses the x-axis (b) $x^2 - 5x + 2 = -x + 3$ draw $y = -x + 3$	**(a)** $x = 0.4$ or 4.6 **(b)** $x = -0.2$ or 4.2 **(line** $y = -x + 3$**)**
1.4	(a) crosses the x-axis (b) $x^2 + x - 4 = x - 2$ draw $y = x - 2$	**(a)** $x = -2.6$ or 1.6 **(b)** $x = -1.4$ or 1.4 **(line** $y = x - 2$**)**
solving with a cubic graph		
1.5	(a) draw $y = -2$ (b) between the turning points	**(a)** $-0.7,\ 1,\ 2.7$ **(b)** $-4 < k < 0$
1.6	(a) draw $y = 1$ (b) between the turning points	**(a)** $-0.5,\ 0.7,\ 2.9$ **(b)** $-2 < k < 2$
1.7	(a) draw $y = -1$ (b) between the turning points	**(a)** $-1.5,\ -0.3,\ 1.9$ **(b)** $-2 < k < 2$
1.8	(a) draw $y = 1$ (b) between the turning points	**(a)** $-1.7,\ 0,\ 1.7$ **(b)** $-1 < k < 3$

Graphs 6 · Non-linear simultaneous equations (Lesson 8.3)

Q	Working	Answer
a line and a circle		

2.1	$x^2 + (x-1)^2 = 13$ $x^2 + x^2 - 2x + 1 = 13$ $2x^2 - 2x - 12 = 0$ $x^2 - x - 6 = 0$ $(x+2)(x-3) = 0$ $x = -2$ or 3 $y = -3$ or 2	$x = -2, y = -3$ **or** $x = 3, y = 2$
2.2	$x^2 + (2x-5)^2 = 10$ $x^2 + 4x^2 - 20x + 25 = 10$ $5x^2 - 20x + 15 = 0$ $x^2 - 4x + 3 = 0$ $(x-1)(x-3) = 0$ $x = 1$ or 3 $y = -3$ or 1	$x = 1, y = -3$ **or** $x = 3, y = 1$
2.3	$x^2 + (x+3)^2 = 17$ $x^2 + x^2 + 6x + 9 = 17$ $2x^2 + 6x - 8 = 0$ $x^2 + 3x - 4 = 0$ $(x+4)(x-1) = 0$ $x = -4$ or 1 $y = -1$ or 4	$x = -4, y = -1$ **or** $x = 1, y = 4$
2.4	$x^2 + (-x+5)^2 = 13$ $x^2 + x^2 - 10x + 25 = 13$ $2x^2 - 10x + 12 = 0$ $x^2 - 5x + 6 = 0$ $(x-2)(x-3) = 0$ $x = 2$ or 3 $y = 3$ or 2	$x = 2, y = 3$ **or** $x = 3, y = 2$
2.5	$x^2 + (3x-10)^2 = 20$ $x^2 + 9x^2 - 60x + 100 = 20$ $10x^2 - 60x + 80 = 0$ $x^2 - 6x + 8 = 0$ $(x-2)(x-4) = 0$ $x = 2$ or 4 $y = -4$ or 2	$x = 2, y = -4$ **or** $x = 4, y = 2$

2.6	$x^2 + (x-2)^2 = 10$ $x^2 + x^2 - 4x + 4 = 10$ $2x^2 - 4x - 6 = 0$ $x^2 - 2x - 3 = 0$ $(x+1)(x-3) = 0$ $x = -1$ or 3 $y = -3$ or 1	$x = -1, y = -3$ **or** $x = 3, y = 1$
a line and a circle on a graph		
2.7	draw $y = x - 1$ read the two crossing points	$x = -3, y = -4$ **or** $x = 4, y = 3$
2.8	draw $y = -x + 7$ read the two crossing points	$x = 3, y = 4$ **or** $x = 4, y = 3$
2.9	draw $y = x + 1$ read the two crossing points	$x = -3, y = -2$ **or** $x = 2, y = 3$
2.10	draw $y = -x + 1$ read the two crossing points	$x = -3, y = 4$ **or** $x = 4, y = -3$

Handling Data 5 · Averages from frequency tables (Lesson 9.1)

Q	Working	Answer
grouped data		
3.1	**(a)** **(b)** $(40+1) \div 2 = 20.5$th add: $4,\ 13,\ 28$ **(c)** $5 \times 4 + 15 \times 9 + 25 \times 15 + 35 \times 8 + 45 \times 4 = 990$ $990 \div 40 = 24.75$	**(a)** $20 < t \leq 30$ **(b)** $20 < t \leq 30$ **(c) 24.75 minutes**
3.2	**(a)** **(b)** $(50+1) \div 2 = 25.5$th add: $6,\ 20,\ 40$ **(c)** $10 \times 6 + 30 \times 14 + 50 \times 20 + 70 \times 7 + 90 \times 3 = 2240$ $2240 \div 50 = 44.8$	**(a)** $40 < m \leq 60$ **(b)** $40 < m \leq 60$ **(c) 44.8 g**

3.3	**(a)** **(b)** $(40+1)\div 2 = 20.5\text{th}$ $\text{add: } 3,\ 14,\ 26$ **(c)** $2.5\times 3 + 7.5\times 11 + 12.5\times 12 + 17.5\times 10 + 22.5\times 4 = 505$ $505\div 40 = 12.62$	**(a)** $10 < t \leq 15$ **(b)** $10 < t \leq 15$ **(c) 12.62 minutes**
3.4	**(a)** **(b)** $(40+1)\div 2 = 20.5\text{th}$ $\text{add: } 2,\ 10,\ 24$ **(c)** $5\times 2 + 15\times 8 + 25\times 14 + 35\times 11 + 45\times 5 = 1090$ $1090\div 40 = 27.25$	**(a)** $20 < t \leq 30$ **(b)** $20 < t \leq 30$ **(c) 27.25 minutes**
3.5	**(a)** **(b)** $(40+1)\div 2 = 20.5\text{th}$ $\text{add: } 7,\ 20,\ 29$ **(c)** $2\times 7 + 6\times 13 + 10\times 9 + 14\times 6 + 18\times 5 = 356$ $356\div 40 = 8.9$	**(a)** $4 < t \leq 8$ **(b)** $8 < t \leq 12$ **(c) 8.9 seconds**
3.6	**(a)** **(b)** $(50+1)\div 2 = 25.5\text{th}$ $\text{add: } 5,\ 15,\ 30$ **(c)** $5\times 5 + 15\times 10 + 25\times 15 + 35\times 12 + 45\times 8 = 1330$ $1330\div 50 = 26.6$	**(a)** $20 < a \leq 30$ **(b)** $20 < a \leq 30$ **(c) 26.6 years**
ungrouped data		
3.7	**(a)** $0\times 3 + 1\times 6 + 2\times 5 + 3\times 4 + 4\times 2 = 36$ $36\div 20 = 1.8$ **(b)** $(20+1)\div 2 = 10.5\text{th}$ $\text{add: } 3,\ 9,\ 14,\ 18,\ 20$	**(a) 1.8** **(b) 2**
3.8	**(a)** $1\times 4 + 2\times 7 + 3\times 5 + 4\times 3 + 5\times 1 = 50$ $50\div 20 = 2.5$ **(b)** $(20+1)\div 2 = 10.5\text{th}$ $\text{add: } 4,\ 11,\ 16,\ 19,\ 20$	**(a) 2.5** **(b) 2**

3.9	**(a)** $0 \times 5 + 1 \times 8 + 2 \times 4 + 3 \times 3 = 25$ $25 \div 20 = 1.25$ **(b)** $(20 + 1) \div 2 = 10.5\text{th}$ add: $5,\ 13,\ 17,\ 20$	**(a) 1.25** **(b) 1**
3.10	**(a)** $0 \times 2 + 1 \times 5 + 2 \times 8 + 3 \times 3 + 4 \times 2 = 38$ $38 \div 20 = 1.9$ **(b)** $(20 + 1) \div 2 = 10.5\text{th}$ add: $2,\ 7,\ 15,\ 18,\ 20$	**(a) 1.9** **(b) 2**
3.11	**(a)** $1 \times 6 + 2 \times 9 + 3 \times 3 + 4 \times 2 = 41$ $41 \div 20 = 2.05$ **(b)** $(20 + 1) \div 2 = 10.5\text{th}$ add: $6,\ 15,\ 18,\ 20$	**(a) 2.05** **(b) 2**
3.12	**(a)** $0 \times 1 + 1 \times 4 + 2 \times 6 + 3 \times 5 + 4 \times 3 + 5 \times 1 = 48$ $48 \div 20 = 2.4$ **(b)** $(20 + 1) \div 2 = 10.5\text{th}$ add: $1,\ 5,\ 11,\ 16,\ 19,\ 20$	**(a) 2.4** **(b) 2**

Graphs 6 · Non-linear simultaneous equations (Lesson 8.3)

Q	Working	Answer
a line and a quadratic		
4.1	$x^2 - 1 = x + 5$ $x^2 - x - 6 = 0$ $(x + 2)(x - 3) = 0$ $x = -2$ or 3 $y = 3$ or 8	$x = -2, y = 3$ **or** $x = 3, y = 8$
4.2	$x^2 = 2x + 3$ $x^2 - 2x - 3 = 0$ $(x + 1)(x - 3) = 0$ $x = -1$ or 3 $y = 1$ or 9	$x = -1, y = 1$ **or** $x = 3, y = 9$

4.3	$x^2 - 2x + 4 = 3x - 2$ $x^2 - 5x + 6 = 0$ $(x - 2)(x - 3) = 0$ $x = 2$ or 3 $y = 4$ or 7	$x = 2, y = 4$ **or** $x = 3, y = 7$
4.4	$x^2 - 2 = -x + 4$ $x^2 + x - 6 = 0$ $(x + 3)(x - 2) = 0$ $x = -3$ or 2 $y = 7$ or 2	$x = -3, y = 7$ **or** $x = 2, y = 2$
4.5	$x^2 + x - 3 = 2x - 1$ $x^2 - x - 2 = 0$ $(x + 1)(x - 2) = 0$ $x = -1$ or 2 $y = -3$ or 3	$x = -1, y = -3$ **or** $x = 2, y = 3$
4.6	$x^2 = 4x - 3$ $x^2 - 4x + 3 = 0$ $(x - 1)(x - 3) = 0$ $x = 1$ or 3 $y = 1$ or 9	$x = 1, y = 1$ **or** $x = 3, y = 9$